

# Limei Wang

Research Scientist, Meta

E-mail: [limeiwang.work@gmail.com](mailto:limeiwang.work@gmail.com), Homepage: <https://limei0307.github.io/> Latest update: December 2025

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## Education

- **Texas A&M University**, College Station, TX, USA *August 2019 - May 2024*  
Ph.D. in Computer Engineering, Department of Computer Science and Engineering  
Advisor: [Prof. Shuiwang Ji](#)
- **University of Science and Technology of China**, Hefei, CHINA *August 2015 - July 2019*  
B.E. in Automation, School of Information Science and Technology  
Advisor: [Prof. Jiahu Qin](#)

## Experience

- Research Scientist, Meta, Bay Area, CA, USA *February 2024 - present*
- Research Intern, Meta, Bay Area, CA, USA *May 2023 - August 2023*  
Work on scalable machine learning methods for large-scale graphs, design a subgraph sampling method  
Mentor: [Dr. Si Zhang](#), Peers: Hanqing Zeng, Hao Wu, Zhigang Hua, Kaveh Hassani

## Research Interests

- Artificial Intelligence, Machine Learning, Deep Learning
- AI for Science, Graph/Geometric ML for Molecules [Related papers: [SphereNet](#), [ComENet](#), [ProNet](#), [LEFT-Net](#), [LatentDiff](#), [Geo2Seq](#), [AIRS](#)]
- Large-Scale Graphs, Graph Transformer, Foundation Models [Related papers: [GraphFM](#), [S3](#), [GQT](#)]
- Generative Models [Related papers: [LatentDiff](#)]
- Large Language Models [Related papers: [Geo2Seq](#), [Haystack Engineering](#)]

## Publications[[Google Scholar](#)]

\* indicates equal contribution. † indicates equal senior contribution.

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|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [NeurIPS-W 2025]                            | <b>Haystack Engineering: Context Engineering for Heterogeneous and Agentic Long-Context Evaluation</b><br>Mufei Li, Dongqi Fu, <b>Limei Wang</b> , Si Zhang, Hanqing Zeng, Kaan Sancak, Ruizhong Qiu, Haoyu Wang, Xiaoxin He, Xavier Bresson, Yinglong Xia, Chonglin Sun, Pan Li                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| [NeurIPS-W 2025, IEEE BigData 2025]         | <b>Staleness-based Subgraph Sampling for Training GNNs on Large-Scale Graphs</b><br><b>Limei Wang</b> , Si Zhang, Hanqing Zeng, Hao Wu, Zhigang Hua, Kaveh Hassani, Andrey Malevich, Bo Long, Shuiwang Ji                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Foundations and Trends® in Machine Learning | <b>Artificial Intelligence for Science in Quantum, Atomistic, and Continuum Systems</b><br>Xuan Zhang*, <b>Limei Wang*</b> , Jacob Helwig*, Youzhi Luo*, Cong Fu*, Yaochen Xie*, Meng Liu, Yuchao Lin, Zhao Xu, Keqiang Yan, Keir Adams, Maurice Weiler, Xiner Li, Tianfan Fu, Yucheng Wang, Haiyang Yu, YuQing Xie, Xiang Fu, Alex Strasser, Shenglong Xu, Yi Liu, Yuanqi Du, Alexandra Saxton, Hongyi Ling, Hannah Lawrence, Hannes Stärk, Shurui Gui, Carl Edwards, Nicholas Gao, Adriana Ladera, Tailin Wu, Elyssa F. Hofgard, Aria Mansouri Tehrani, Rui Wang, Ameya Daigavane, Montgomery Bohde, Jerry Kurtin, Qian Huang, Tuong Phung, Minkai Xu, Chaitanya K. Joshi, Simon V. Mathis, Kamyar Azizzadenesheli, Ada Fang, Alán Aspuru-Guzik, Erik Bekkers, Michael Bronstein, Marinka Zitnik, Anima Anandkumar, Stefano Ermon, Pietro Liò, Rose Yu, Stephan Günnemann, Jure Leskovec, Heng Ji, Jimeng Sun, Regina Barzilay, Tommi Jaakkola, Connor W. Coley, Xiaoning Qian, Xiaofeng Qian, Tess Smidt, Shuiwang Ji |

- [ICML 2025] **Geometry Informed Tokenization of Molecules for Language Model Generation**  
Xiner Li\*, Limei Wang\*, Youzhi Luo, Carl Edwards, Shurui Gui, Yuchao Lin, Heng Ji, Shuiwang Ji
- [ICLR 2025] **Learning Graph Quantized Tokenizers**  
Limei Wang\*, Kaveh Hassani\*, Si Zhang, Dongqi Fu, Baichuan Yuan, Weilin Cong, Zhigang Hua, Hao Wu, Ning Yao, Bo Long
- [LoG 2023] **A Latent Diffusion Model for Protein Structure Generation**  
Cong Fu\*, Keqiang Yan\*, Limei Wang, Wing Yee Au, Michael McThrow, Tao Komikado, Koji Maruhashi, Kanji Uchino, Xiaoning Qian, Shuiwang Ji
- [NeurIPS 2023] **A New Perspective on Building Efficient and Expressive 3D Equivariant Graph Neural Networks**  
Weitao Du\*, Yuanqi Du\*, Limei Wang\*, Dieqiao Feng, Guifeng Wang, Shuiwang Ji, Carla P Gomes, and Zhi-Ming Ma
- [ICLR 2023] **Learning Hierarchical Protein Representations via Complete 3D Graph Networks**  
Limei Wang\*, Hanran Liu\*, Yi Liu\*,<sup>†</sup>, Jerry Kurtin, and Shuiwang Ji<sup>†</sup>
- [NeurIPS 2022] **ComENet: Towards Complete and Efficient Message Passing for 3D Molecular Graphs**  
Limei Wang\*, Yi Liu\*, Yuchao Lin, Haoran Liu, and Shuiwang Ji
- [NeurIPS-D&B Track 2022] **GOOD: A Graph Out-of-Distribution Benchmark**  
Shurui Gui\*, Xiner Li\*, Limei Wang, and Shuiwang Ji
- [KDD 2022 Tutorial] **Frontiers of Graph Neural Networks with DIG**  
Shuiwang Ji, Meng Liu, Yi Liu, Youzhi Luo, Limei Wang, Yaochen Xie, Zhao Xu, Haiyang Yu
- [ICML 2022] **GraphFM: Improving Large-Scale GNN Training via Feature Momentum**  
Haiyang Yu\*, Limei Wang\*, Bokun Wang\*, Meng Liu, Tianbao Yang, and Shuiwang Ji
- [Bioinformatics 2022] **Advanced Graph and Sequence Neural Networks for Molecular Property Prediction and Drug Discovery**  
Zhengyang Wang\*, Meng Liu\*, Youzhi Luo\*, Zhao Xu\*, Yaochen Xie\*, Limei Wang\*, Lei Cai\*, Qi Qi, Zhuoning Yuan, Tianbao Yang, and Shuiwang Ji
- [ICLR 2022] **Spherical Message Passing for 3D Molecular Graphs**  
Yi Liu\*, Limei Wang\*, Meng Liu, Yuchao Lin, Xuan Zhang, Bora Oztekin, and Shuiwang Ji
- [JMLR 2021] **DIG: A Turnkey Library for Diving into Graph Deep Learning Research**  
Meng Liu\*, Youzhi Luo\*, Limei Wang\*, Yaochen Xie\*, Hao Yuan\*, Shurui Gui\*, Haiyang Yu\*, Zhao Xu, Jingtun Zhang, Yi Liu, Keqiang Yan, Haoran Liu, Cong Fu, Bora Oztekin, Xuan Zhang, and Shuiwang Ji
- [NeurIPS-W 2021] **Fast Quantum Property Prediction via Deeper 2D and 3D Graph Networks**  
Meng Liu\*, Cong Fu\*, Xuan Zhang, Limei Wang, Yaochen Xie, Hao Yuan, Youzhi Luo, Zhao Xu, Shenglong Xu, and Shuiwang Ji
- [Chemistry-Methods 2021] **Development of Xanthene-Based Fluorescent Dyes: Machine Learning-Assisted Prediction vs. TD-DFT Prediction and Experimental Validation**  
Yingying Wang\*, Lei Cai\*, Wei Chen, Difei Wang, Shi Xu, Limei Wang, Martin A. Kononov, Shuiwang Ji, and Ming Xian

## Preprints

### **How to Make LLMs Strong Node Classifiers?**

Zhe Xu, Kaveh Hassani, Si Zhang, Hanqing Zeng, Michihiro Yasunaga, **Limei Wang**, Dongqi Fu, Ning Yao, Bo Long, Hanghang Tong

## Professional Services

### **Program Committee Member & Reviewer**

- International Conference on Machine Learning (**ICML**) 2022, 2023, 2024
- Conference on Neural Information Processing Systems (**NeurIPS**) 2022, 2023
- NeurIPS Datasets and Benchmarks Track 2022, 2023
- AI4Science Workshop @ NeurIPS 2022, 2023
- International Conference on Learning Representations (**ICLR**) 2023, 2024
- Learning on Graphs Conference (**LoG**) 2022, 2023, 2025
- ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**) 2022

## Talks, Lecture, & Tutorials

### **Invited Talks**

- Geometric Deep Learning for 3D Molecules *Lawrence Livermore National Laboratory, 2024*
- Improving Large-Scale GNN Training via Feature Momentum *Meta, 2023*
- Learning Hierarchical Protein Representations via 3D Graph Networks *Fujitsu Research of America, 2022*

### **Tutorial**

- PyTorch Tutorial *CSCE 421, Texas A&M University, 2023*
- [Frontiers of Graph Neural Networks with DIG](#) *Texas A&M Institute of Data Science, 2023*  
*KDD, 2022*

## Scholarships, Awards, & Honors

- [EECS Rising Star](#), Georgia Tech 2023
- NeurIPS Travel Award 2022, 2023
- Travel Grant, CSE@TAMU 2021, 2022, 2023
- Third place on the [Open Catalyst Challenge](#) 2021
- Runner-Up Award of KDD Cup on [Open Graph Benchmark Large-Scale Challenge \(OGB-LSC\)](#) 2021
- 1st Place, MIT AI Cures Challenge for COVID-19 Drug Discovery 2020
- Outstanding Graduates, USTC 2019

## News Coverage

- [Ph.D. Student Attends Workshop for Rising Stars](#) *TAMU News 2024*
- [Ji and his team earn top showing at premier data mining competition](#) *TAMU News 2021*
- [Computer scientists using artificial intelligence to fight secondary effects of COVID-19](#) *TAMU News 2020*